Assumption 1 The Markov process generated by the SDE (6.43) satisfies, for some fixed compact set $C \in \mathcal{B}(\mathcal{D})$, the conditions:

(i) For some $\boldsymbol{y} \in \text{int}(C)$, we have that for any $\delta > 0$ there is a $t_0 > 0$ such that

$$\Pr\left(\mathbf{X}(t_0) \in \mathcal{B}_\delta(\boldsymbol{y}) \,|\, \mathbf{X}(0) = \mathbf{X}_0\right) > 0, \quad \forall \mathbf{X}_0 \in C.$$

(ii) For $t > 0$, the transition kernel possesses a density $\hat{\rho}(\mathbf{X}, \mathbf{z}, t)$ such that

$$\Pr\left(\mathbf{X}(t) \in A \,|\, \mathbf{X}(0) = \mathbf{X}_0\right) = \int_A \hat{\rho}(\mathbf{X}_0, \mathbf{z}, t) \mathrm{d}\mathbf{z}, \quad \forall \mathbf{X}_0 \in C,\ A \in \mathcal{B}(\mathcal{D}) \cap \mathcal{B}(C),$$

and $\hat{\rho}(\mathbf{X}, \mathbf{z}, t)$ is jointly continuous in $C \times C \times [0, \infty)$.

It can be shown that this assumption is tantamount to the SDE satisfying a *minorization condition*. This condition is a particular case of the general concept of small sets, found in Markov literature for proving ergodicity. We can think of it as giving conditions on the regularity of the solution, in some small region C. The return to the set C is governed by the following assumption:

Assumption 2 There is a function $\phi : \mathcal{D} \to (0, \infty)$, with $\phi(\mathbf{X}) \to \infty$ as $|\mathbf{X}| \to \infty$ such that

$$\mathcal{L}\phi \leq -\alpha\phi + \delta$$

for positive constants α and δ, where $\mathcal{L}$ is the generator of the stochastic process (6.43). The function ϕ is known as a Lyapunov function.

If these two assumptions hold, then we are able to state the following theorem [165]:

Theorem 6.2 (Geometric Ergodicity) *Let the solution to* (6.43) *be denoted* $\mathbf{X}(t)$. *If Assumption 1 and Assumption 2 are satisfied for*

$$C = \{\mathbf{X} \in \mathcal{D} \,|\, \phi(\mathbf{X}) \leq K_{\alpha,\delta}\},$$

where $K_{\alpha,\delta}$ *is a particular positive value depending on* α *and* δ, *then*

(i) there exists a unique invariant measure ρ_*, *with* $\mathcal{L}^\dagger \rho_* = 0$, *and*
(ii) there exist positive constants κ *and* λ *such that for all suitable functions* f *with* $|f| \leq \phi$,

$$\left|\left(e^{t\mathcal{L}} f\right)(\mathbf{X}_0) - \text{Av}(f)\right| \leq \kappa e^{-\lambda t} \phi(\mathbf{X}_0), \tag{6.48}$$

where $\text{Av}(f) = \int_{\mathcal{D}} f(\mathbf{z}) \rho_*(\mathbf{z}) \,\mathrm{d}\mathbf{z}$ *is the spatial average.*

It is easily shown that both the Ornstein-Uhlenbeck equation and Brownian dynamics satisfy the assumptions and hence are both geometrically ergodic. We will show Langevin dynamics is geometrically ergodic by applying Theorem 6.2